Macroeconomics I — MPE Insper

Practice list: Consumption, Saving and the Labor Market (Listas 3–4)

Solutions and grading rubric

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Numbers are illustrative and chosen to give clean answers.

Block 1 — Consumption and saving

(Kurlat ch. 6 • 25 points)

Question 1. (10 points) Objective items. A young worker in São Paulo lives two periods, with $u(c_1) + \beta u(c_2)$, $u = \ln c$, $\beta = 0.8$, income $y_1 = 60$, $y_2 = 150$, and borrows or lends freely at $r = 0.25$ (illustrative; a period is a decade).

a) (3 points) The worker's optimal saving $a = y_1 - c_1$ is:

A. -30

B. $+40$

C. -40

D. -20

Answer: C

Step — wealth. $W = y_1 + y_2/(1+r) = 60 + 150/1.25 = 180$.

Step — log rule. $c_1 = W/(1+\beta) = 180/1.8 = 100$, so $a = 60 - 100 = -40$: the worker borrows against high future income.

Ref: Kurlat (2020), §6.2, p. 105.

b) (3 points) Replace log utility by $u(c) = c^{1-\sigma}/(1-\sigma)$ with $\sigma = 2$, keeping β, r and incomes. Then c_1 equals:

A. 120.0

B. 100.0

C. 90.0

D. 108.3

Answer: B

Step — Euler. $c_2/c_1 = [\beta(1+r)]^{1/\sigma} = 1^{1/2} = 1$, so $c_1 = c_2 = c$ and $c(1 + 1/1.25) = 180$ gives $c = 100$. When $\beta(1+r) = 1$ the household smooths perfectly whatever its EIS $1/\sigma$.

Ref: Kurlat (2020), §6.2.

c) (2 points, T or F) For the worker in (a), a tax cut of 10 in period 1 financed by a tax increase of 12.5 in period 2 raises c_1.

Answer: F

Step — Ricardian equivalence. Present value of taxes changes by $-10 + 12.5/1.25 = 0$, so W and consumption are unchanged; saving rises by 10.

Ref: Kurlat (2020), §6.2 (taxes) and §6.3.

d) (2 points, T or F) If the worker faces the borrowing limit $a \geq -10$, the same tax change raises c_1 by exactly 10.

Answer: T

Step — binding limit. Unconstrained $c_1 = 100$ needs $a = -40 < -10$: the limit binds and $c_1 = y_1 + 10 = 70$. After the cut $c_1 = 60 + 10 + 10 = 80$, still below the desired 100, so the limit still binds and c_1 rises one for one.

Ref: Kurlat (2020), §6.2, borrowing constraints.

Question 2. (15 points) Discursive. A worker earns y in period 1 and nothing in period 2 (retirement). Preferences $u(c_1) + \beta u(c_2)$ with $u(c) = c^{1-\sigma}/(1-\sigma)$ ($\ln c$ if $\sigma = 1$); the worker saves at rate r.

a) (5 points) Derive c_1, c_2 and a in closed form. Show how the sign of $\partial a/\partial r$ depends on σ, and explain it with income and substitution effects.

Step — problem. $\max u(c_1) + \beta u(c_2)$ s.t. $c_1 + \dfrac{c_2}{1+r} = y$.

Step — Euler. $c_1^{-\sigma} = \beta(1+r)c_2^{-\sigma} \Rightarrow c_2 = [\beta(1+r)]^{1/\sigma} c_1$.

Step — solution. $c_1 = \dfrac{y}{1+\beta^{1/\sigma}(1+r)^{1/\sigma-1}}$, $c_2 = [\beta(1+r)]^{1/\sigma} c_1$, $a = y - c_1$.

Step — sign. a rises with r iff the denominator rises with r, i.e. iff $1/\sigma - 1 > 0$: $\sigma < 1 \Rightarrow \partial a/\partial r > 0$; $\sigma = 1 \Rightarrow 0$; $\sigma > 1 \Rightarrow < 0$.

Economic intuition. A higher r makes future consumption cheaper (substitution: save more) and makes a saver richer (income: consume more today). The EIS $1/\sigma$ sets the strength of substitution; with log the two cancel exactly.

Ref: Kurlat (2020), §6.2, pp. 105–117.

Grading rubric.

- Budget constraint and Euler: 2 pts. Discounting c_2 by multiplying by $(1+r)$: max. 2 pts.
- Closed forms: 1.5 pts.
- Sign by σ with income/substitution explanation: 1.5 pts. Sign stated in words that contradicts the derived exponent: max. 2 pts in the item (words must agree with the math).

b) (5 points) Let $\sigma = 1$, $\beta = 0.8$, $r = 0.25$, $y = 150$. The government taxes interest income at $\tau = 0.4$, so $c_2 = (1 + (1-\tau)r)a + T$, and returns the revenue as a lump-sum T in period 2 (the worker takes T as given). Compute c_1, c_2, a and T with and without the tax. Show that the tax changes c_2/c_1 while an equal-revenue lump-sum tax would not, and measure the welfare cost as the percentage cut in consumption in both periods that would give the untaxed worker the same utility.

Step — no tax. $\beta(1+r) = 1$: $c_1 = c_2 = 150/1.8 = 83.33$, $a = 66.67$.

Step — with tax. Euler uses the after-tax return: $c_2/c_1 = \beta(1 + 0.6 \cdot 0.25) = 0.92$. In equilibrium $T = \tau r a$, so $c_2 = (1+r)a$ and the resource constraint is still $c_1 + c_2/1.25 = 150$: $c_1(1 + 0.92/1.25) = 150 \Rightarrow c_1 = 86.41$, $c_2 = 79.49$, $a = 63.59$, $T = 0.4 \cdot 0.25 \cdot 63.59 = 6.36$.

Step — lump-sum comparison. A lump-sum tax leaves the return at r: Euler $c_2/c_1 = \beta(1+r) = 1$. The interest tax drives a wedge between the private return $(1-\tau)r$ and the social return r: it distorts the intertemporal margin.

Step — welfare. $\Delta U = \ln \frac{86.41}{83.33} + 0.8 \ln \frac{79.49}{83.33} = -0.00154$; the cut λ solves $1.8 \ln(1-\lambda) = -0.00154$: $\lambda = 0.086\%$.

Economic intuition. Log utility makes saving insensitive to r only when income and substitution effects both act. Rebating the revenue removes the income effect and leaves pure substitution, so saving falls (66.67 to 63.59).

Ref: Kurlat (2020), §6.2; Lista 3, Q2(d).

Grading rubric.

- Untaxed allocation: 1 pt.
- Taxed Euler and allocation, revenue: 2 pts. Using r instead of $(1-\tau)r$ in the Euler: max. 1 pt.
- Wedge argument (which margin is distorted): 1 pt.
- Welfare cost: 1 pt.

c) (5 points) Return to the worker of Question 1(a) ($y_1 = 60$, $y_2 = 150$, log, $\beta = 0.8$, $r = 0.25$). Compare the response of c_1 to (i) a one-off transfer of 10 in period 1 (like the 2020 *auxílio emergencial*) and (ii) a permanent rise of 10 in both periods. Redo both for a worker facing $a \geq -10$. What does this say about the permanent income hypothesis and about the Keynesian consumption function?

Step — unconstrained. (i) $\Delta W = 10$, $\Delta c_1 = 10/1.8 = 5.56$, $\Delta a = 4.44$. (ii) $\Delta W = 10 + 10/1.25 = 18$, $\Delta c_1 = 10$, $\Delta a = 0$.

Step — constrained. The limit binds in both cases (desired $c_1 = 105.6$ and 110 exceed $y_1 + 10 + \Delta y_1 = 80$), so $c_1 = y_1 + 10$ and $\Delta c_1 = 10$ in both.

Step — reading. The unconstrained worker responds to wealth, not current income (PIH): the marginal propensity to consume out of a transitory transfer is $1/(1+\beta) = 0.56$. Constrained workers have MPC $= 1$: the Keynesian consumption function reappears exactly where credit fails.

Economic intuition. Aggregate MPC from a one-off transfer depends on how many households are credit-constrained, which is why stimulus checks are more effective in poor, unbanked economies.

Ref: Kurlat (2020), §6.1 (Keynesian) and §6.2–6.3.

Grading rubric.

- Both unconstrained responses: 2 pts.
- Constrained responses with the check that the limit still binds: 2 pts.
- PIH vs Keynesian interpretation: 1 pt.

Block 2 — Labor supply and unemployment *(Kurlat ch. 7 • 25 points)*

Question 3. (10 points) Objective items. A PNAD Contínua respondent has $u = \ln c + \theta \ln l$, $\theta = 1$, one unit of time ($h = 1 - l$), real wage $w = 20$ and non-labor income $\pi = 4$.

a) (3 points) Hours worked are:

A. 0.6

B. 0.4

C. 0.5

D. 0.2

Answer: B

Step — FOC. $\theta c/l = w$ with $c = w(1-l) + \pi$: $l = \frac{\theta}{1+\theta} \cdot \frac{w+\pi}{w} = \frac{1}{2} \cdot \frac{24}{20} = 0.6$, so $h = 0.4$. Option A is leisure; C is the $\pi = 0$ answer.

Ref: Kurlat (2020), §7.2, p. 131.

b) (3 points) The reservation wage (below which she does not work) is:

A. 2

B. 8

C. 20

D. 4

Answer: D

Step — corner. $w^r = u_l/u_c$ at $(c = \pi,\ l = 1)$: $w^r = \theta\pi/1 = 4$.

Ref: Kurlat (2020), §7.2.

c) (2 points, T or F) For this respondent, a 10% rise in the after-tax wage leaves hours unchanged, because with log utility income and substitution effects cancel.

Answer: F

Step — check. $h = 1 - \frac{1}{2}(1 + \pi/w)$ rises with w when $\pi > 0$. The cancellation needs $\pi = 0$: non-labor income weakens the income effect of a wage rise.

Ref: Kurlat (2020), §7.2.

d) (2 points, T or F) In the DMP model with $m = \mu V^{\alpha} U^{1-\alpha}$, a rise in matching efficiency μ shifts the Beveridge curve toward the origin: fewer vacancies are needed at each unemployment rate.

Answer: T

Step — Beveridge. Steady state: $s(1-u) = \mu v^{\alpha} u^{1-\alpha}$. At given u, higher μ needs lower v to create the same flow of matches.

Ref: Kurlat (2020), §7.5, p. 142; Lista 4, Q2.

Question 4. (15 points) Discursive. Brazil's labor tax wedge. A household has $U = \ln c + b\,\dfrac{l^{1-\gamma}}{1-\gamma}$ $(b, \gamma > 0)$, one unit of time, wage w, labor-income tax τ and a lump-sum transfer T: $c = (1-\tau)w(1-l) + T$.

a) (5 points) Derive the labor supply condition. Show that when $T = 0$ hours depend on neither w nor τ, for any γ. Show that when $T > 0$ hours rise with $(1-\tau)w$.

Step — FOC. $b\,l^{-\gamma} = \dfrac{(1-\tau)w}{c} = \dfrac{(1-\tau)w}{(1-\tau)w(1-l)+T}$.

Step — $T = 0$. $b\,l^{-\gamma}(1-l) = 1$: w and τ drop out. Income and substitution effects of the net wage cancel.

Step — $T > 0$. Rewrite as $G(l) \equiv b\,l^{-\gamma}\big[(1-l) + \frac{T}{(1-\tau)w}\big] = 1$, with $G'(l) < 0$. A higher $(1-\tau)w$ lowers the bracket; to restore $G = 1$, l must fall: hours rise.

Economic intuition. The transfer is income that does not scale with the wage, so a wage rise has a smaller income effect than substitution effect.

Ref: Kurlat (2020), §7.2; Lista 4, Q1.

Grading rubric.

- FOC: 2 pts.
- $T = 0$ result: 1.5 pts.
- $T > 0$ comparative statics with a sign argument: 1.5 pts. Asserting the sign without proof: 0.5 pt.

b) (5 points) Now the budget balances: $T = \tau w(1-l)$ in equilibrium (each household still takes T as given). With $\gamma = 1$, $b = 1$, find hours as a function of τ, compute them for $\tau = 0$ and $\tau = 0.5$ and $dh/d\tau$ at $\tau = 0$. Why is the effect larger than in (a) with $T = 0$? Relate to Prescott's comparison of hours in Europe and the US.

Step — equilibrium consumption. $c = (1-\tau)w(1-l) + \tau w(1-l) = w(1-l)$.

Step — hours. FOC: $1/l = (1-\tau)/(1-l) \Rightarrow l = 1/(2-\tau)$, $h = (1-\tau)/(2-\tau)$: $h(0) = 0.5$, $h(0.5) = 1/3$; $dh/d\tau = -1/(2-\tau)^2 = -0.25$ at $\tau = 0$.

Step — why. The rebate gives back the income, so the income effect vanishes and only substitution remains: the tax works like a pure fall in the price of leisure.

Step — Prescott. Higher effective labor taxes in Europe, with revenue returned as transfers, can explain much of its lower hours, under elasticities of this size.

Economic intuition. Taxes on labor depress hours through what they finance as well as what they take: spending returned lump-sum keeps consumption up while the price of leisure stays low.

Ref: Kurlat (2020), §7.3 (evidence, Prescott 2004).

Grading rubric.

- Uses $T = \tau w(1-l)$ only after the FOC (household takes T as given): 1.5 pts. Substituting T before differentiating: max. 2 pts in the item.
- $h(\tau)$, numbers and derivative: 2 pts.
- Income-effect argument and Prescott: 1.5 pts.

c) (5 points) DMP model: separation rate $s = 0.03$ per month, $m = \mu V^{0.5} U^{0.5}$, $\mu = 0.6$, tightness $\theta = V/U = 0.25$. Compute f, q, u^* and v^*. After the pandemic, mismatch lowers μ to 0.45. Compute u^* at the same θ and the vacancy rate needed to keep u at its old value. Explain the shift versus a movement along the Beveridge curve, and why a firm posting one more vacancy creates an externality.

Step — rates. $f = \mu\theta^{0.5} = 0.3$, $q = \mu\theta^{-0.5} = 1.2$ (check $f = \theta q$).

Step — steady state. $s(1-u) = fu \Rightarrow u^* = s/(s+f) = 0.03/0.33 = 9.09\%$; $v^* = \theta u^* = 2.27\%$.

Step — lower μ. $f = 0.45 \cdot 0.5 = 0.225$, $u^* = 0.03/0.255 = 11.76\%$. To keep $u = 9.09\%$: $v = \left(s(1-u)/(\mu u^{0.5})\right)^2 = 4.04\%$, almost twice as many vacancies.

Step — shift vs movement. A fall in μ moves the whole curve away from the origin (more of both u and v). A demand change moves the economy along a fixed curve (v and u in opposite directions).

Step — externality. One more vacancy raises θ: f rises (helps workers), q falls (hurts other firms). The firm weighs only its own expected match, so these effects are not priced.

Ref: Kurlat (2020), §7.5; Lista 4, Q2.

Grading rubric.

- f, q, u^*, v^*: 1.5 pts.
- New u^* and required v: 1.5 pts.
- Shift vs movement, with axes (u, v) named: 1 pt.
- Externality with both sides named: 1 pt.